

ECO-SCOPE

Environmental Compliance Intelligence Platform

THE IDEA IN ONE SENTENCE

Turn Environmental Clearance documents into machine-checkable rules, compare those rules with satellite observations, and flag possible non-compliance for human review.

18	8	24
problems identified	major upgrades	month roadmap

A redesigned, easier-to-understand version of the uploaded ECO-SCOPE build document.

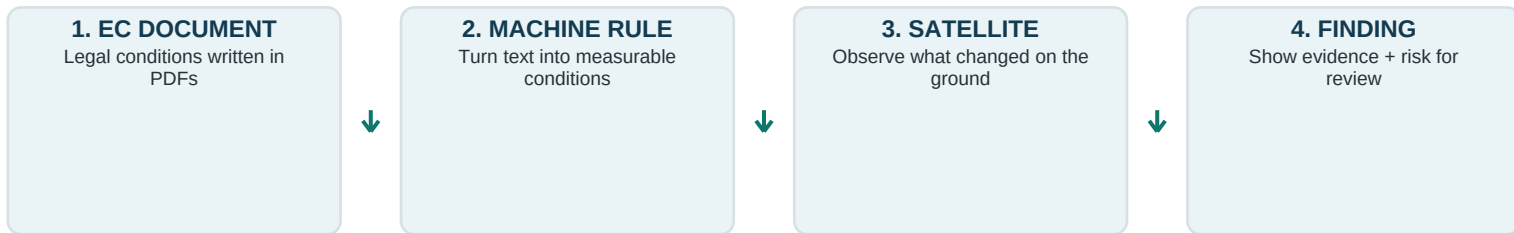
Source basis: the uploaded 22-page ECO-SCOPE Build Roadmap / upgraded specification. The structure and wording below are simplified for understanding; the underlying proposed architecture, risks, roadmap and targets come from the source document.

What is ECO-SCOPE?

The simplest way to understand the product.

ECO-SCOPE is not meant to replace an environmental inspector or a legal decision-maker. It is a **screening and monitoring system** that helps a human reviewer find where a project may be violating conditions in its Environmental Clearance (EC).

The core trick is to connect four things that normally live separately:



Example

EC says: “No construction within 500 m of the river.” ECO-SCOPE tries to identify the river, build a 500 m buffer, detect construction inside that area, and produce an evidence package. It does not declare the project legally guilty.

Why this is valuable

- Large projects generate complex EC documents that are difficult to monitor manually.
- Satellite imagery gives repeated observations without requiring a site visit every time.
- The system creates a traceable link between the original EC condition, the measurement, the satellite evidence and the reviewer’s decision.

Why the problem is harder than it looks

The original concept was strong, but the source document identified 18 technical and product problems.

Documents	Scanned PDFs, tables, Hindi/regional text, stamps and poor OCR.	Bad extraction creates bad rules.
Location	ECs often use village names, survey numbers and landmarks instead of coordinates.	The project boundary may not be directly available.
Satellite data	Clouds, seasons and resolution limits affect what can be observed.	Naive comparisons create false alarms.
Rules	Conditions can be vague: “adequate green cover”, “appropriate distance”.	Some conditions cannot be checked automatically.
Ground truth	Mining-specific labeled data is limited.	Models cannot be trusted without expert validation.
Compliance logic	Approved activity can look like environmental damage.	The system needs expected vs unexpected change.
Operations	One-time analysis misses changes after the report is generated.	Compliance is continuous.
Trust	Confidence and risk scores can be misleading if their calculation is hidden.	Users need to understand why a finding is risky.

Most important technical risk

The source document calls out boundary resolution as a major risk: an EC may say “Survey Nos. 45/1–52/3 of village X” rather than provide a polygon. ECO-SCOPE therefore needs automated lookup where possible and a human map-based fallback when it cannot resolve the boundary.

How ECO-SCOPE works

The end-to-end pipeline, without the jargon.

01

Upload EC

User uploads the Environmental Clearance PDF and amendments.

02

Read it

Document AI cleans the PDF, reads text/tables and extracts conditions.

03

Resolve location

Village, survey numbers, coordinates and water bodies are turned into map geometry.

04

Create rules

A condition becomes a measurable rule with thresholds and a required data resolution.

05

Observe

Sentinel-1, Sentinel-2, Landsat and, only when needed, higher-resolution imagery are searched.

06

Compare

Season, clouds, time gaps and resolution are considered before detecting change.

07

Score

The system calculates evidence strength and risk, with the contributing factors visible.

08

Review

A human confirms, rejects or marks the finding inconclusive.

09

Learn

Verified results become feedback for improving extraction and classification.

What the AI actually does

AI is used in several different places — not as one magical model.

Document understanding	Scanned EC PDFs	Clean text, tables, condition blocks and extraction confidence
Condition extraction	EC language	Condition type, numbers, references and spatial meaning
Ambiguity detection	Qualitative conditions	Precise / Referenceable / Qualitative / Uncheckable
Land-cover classification	Satellite bands + indices	Vegetation, built-up, bare soil, water, mining classes
Change detection	Historical + current imagery	Where land cover or structures changed
Active learning	Human verification results	New training examples for uncertain cases

Important: the source specification proposes starting with foundation models / existing datasets, then building mining-specific training data. It specifically calls for expert labeling of roughly 50–100 mining sites and an active-learning loop.

The human is still in the loop

If the model cannot confidently interpret a condition or location, ECO-SCOPE is designed to ask a reviewer rather than silently invent an answer.

How satellite monitoring avoids false alarms

This is where the upgraded design becomes much more realistic.

Problem	ECO-SCOPE approach
Monsoon clouds	Use Sentinel-1 SAR because radar can operate through clouds; use cloud-masked composites when optical scenes are limited.
Seasonal vegetation	Compare similar seasons and use a 3–5 year same-season baseline rather than a single before/after image.
Different condition sizes	Use resolution tiers: Sentinel-2 for broad changes, finer imagery for boundary/structure questions, and very high resolution only for targeted verification.
Uncertain measurements	Carry classification, boundary and image-quality uncertainty into the final measurement.

Three resolution tiers

- **Tier 1:** broad-area green cover and large land-use change — Sentinel-2 is suitable.
- **Tier 2:** buffer encroachment and construction near boundaries — finer analysis may be required.
- **Tier 3:** small structures or precise boundary questions — trigger high-resolution imagery only for that finding.

Key design principle

Free, frequent satellite data handles routine screening. Paid high-resolution imagery is reserved for cases where a more precise answer is actually needed.

How a compliance finding is scored

The source replaces an unexplained “HIGH/MEDIUM/LOW” with transparent components.

First, ECO-SCOPE estimates **how much the observed situation differs from the requirement**. Then it estimates **how strong the evidence is**. The source proposes:

Risk = Change Magnitude × Evidence Strength / 100

Example from the source: a magnitude score of 72 and evidence strength of 78 gives a risk score of 56, which is HIGH under the proposed thresholds.

Change Magnitude	How far the observed measurement is from the EC requirement.
Classification confidence	How reliable the land-cover model is for that class and region.
Temporal confidence	Whether the comparison dates are close to the ideal same-season window.
Spatial confidence	Whether the imagery resolution is adequate for the condition.
Cloud / image quality	How much cloud contamination or temporal gaps affect the evidence.
Rule confidence	How confidently the EC condition itself was extracted and parameterized.

Why this matters

A reviewer should be able to see not just “HIGH RISK”, but why: large deviation, strong classification, poor temporal match, weak boundary confidence, etc.

What the user sees

The product is designed around evidence, not a black-box score.

Screen	Purpose
Project dashboard	Current risk, latest analysis, trend and project status.
Map view	Project boundary, buffers, detected changes, approved activity zones and evidence layers.
Finding review	Condition → measurement → map evidence → confidence → risk → reviewer decision.
Portfolio view	Compare many projects by risk, geography, trend and review priority.
Evidence report	A watermarked PDF containing the EC condition, maps, measurements, timeline and confidence factors.

The report should always say that it is automated screening and requires human verification. The source proposes persistent watermarks, mandatory acknowledgment before download, authenticated access, embedded metadata and immutable report records.

The 8 upgrades that turn the idea into a real system

These are the major upgrades proposed beyond the original concept.

U01

SAR monitoring

Use Sentinel-1 radar to keep monitoring during cloudy months.

U02

Seasonal normalization

Compare vegetation against a multi-year, same-season baseline.

U03

Active learning

Use verified findings to improve the models over time.

U04

Continuous monitoring

Automatically re-check projects and alert reviewers when new evidence appears.

U05

PARIVESH integration

Use the official EC portal as a metadata / amendment cross-check, subject to access.

U06

Multi-resolution fusion

Use free imagery routinely and paid high-resolution imagery only when needed.

U07

Uncertainty propagation

Show confidence intervals instead of pretending measurements are exact.

U08

Ambiguity handler

Do not silently drop vague EC conditions; route them to human parameterization.

24-month roadmap — simplified

The source roadmap is six phases. For a hackathon, only the early proof matters.

1. Foundation	1–3	EC ingestion + OCR + condition extraction	≥85% extraction recall on held-out ECs
2. Core engine	4–6	Boundary resolution + rules + satellite pipeline	5 mining ECs analyzed end-to-end
3. Intelligence	7–9	Risk scoring + evidence reports + SAR + seasonal baseline	10 complete reports reviewed by consultants
4. Product	10–12	Auth, projects, amendments, verification, portfolio dashboard	5 beta users; NPS ≥30
5. Scale	13–18	Continuous monitoring + active learning + integrations	200+ monitored projects; <30 min/project; FP <25%
6. Advanced	19–24	Forecasting, cross-project patterns, government/API features	Third-party audit; FP <15%; government pilot

If this is a hackathon

Do not build the 24-month platform. Prove one narrow loop: upload an EC → extract 1–3 conditions → map one project → run satellite change detection → show an evidence-backed finding. The full roadmap can remain the scaling plan.

Hackathon MVP — what to actually build

A much smaller version can demonstrate the core insight.

1**Pick one domain**

Mining projects only.

2**Pick three condition types**

Green-cover percentage, project boundary, river/buffer restriction.

3**Use a controlled document set**

A small set of EC PDFs with manual correction available.

4**Use one satellite stack**

Sentinel-2 for optical analysis; Sentinel-1 only where cloud cover makes it necessary.

5**Use human-in-the-loop boundaries**

Do not spend the hackathon trying to solve every Indian cadastral database.

6**Show the evidence**

Before/after imagery, detected change, EC condition, measurement and confidence.

7**End with review**

Confirmed / Rejected / Inconclusive.

Demo story

“This EC required 33% green cover. The system extracted that condition, mapped the project, compared same-season imagery, estimated current green cover, and showed the evidence and uncertainty. A reviewer can now confirm or reject the finding.”

Technical architecture

Six layers keep the system understandable and modular.

1	Document intelligence	PDF → preprocessing → layout analysis → OCR → condition extraction → ambiguity
2	Geospatial resolution	Text locations → geocoding/cadastral lookup → boundary → water body → validated geometry
3	Rule generation	Conditions → rule schema → resolution tier → human parameterization → validated rules
4	Earth observation	AOI + time → Sentinel/Landsat search → cloud/season filter → classification/change → uncertainty
5	Compliance engine	Rules + maps → spatial tests → confidence → risk → findings
6	Output + feedback	Findings → evidence package → watermarked report → verification → training data

Source technology choices

Frontend: React + TypeScript + MapLibre GL JS | **Backend:** Python/FastAPI | **Document AI:** LayoutLMv3 + PaddleOCR | **Geospatial:** Google Earth Engine + rasterio/GDAL | **ML:** U-Net / Random Forest starting point | **Database:** PostgreSQL + PostGIS + Redis | **Queue:** Celery + Redis | **Storage:** S3 / GCP Cloud Storage

Risks and success metrics

The source document is explicit about what could break the system.

Cadastral data unavailable	High	Human map drawing fallback; prioritize states with digitized records.
Mining-class accuracy too low	High	More expert labels; use SAR as complementary evidence.
False positives stay high	Medium	Active learning + targeted training on failure modes.
PARIVESH access changes	Low	Manual EC upload remains the fallback.
Legal challenge	High	Watermarks, restricted sharing, legal review and clear positioning.
Users misunderstand “low risk”	Medium	Use wording such as “low risk of detected non-compliance” and keep disclaimers visible.

24-month target snapshot

Metric	Month 6	Month 12	Month 24
EC extraction recall	≥85%	≥92%	≥95%
Spatial non-compliance detection	≥70%	≥80%	≥85%
False positive rate	≤40%	≤25%	≤15%
Boundary auto-resolution	≥40%	≥60%	≥70%
End-to-end latency	≤60 min	≤30 min	≤15 min
Active projects	10	75	300+

Bottom line

What the whole document is really saying.

The core idea

Environmental Clearance conditions are written for humans. Satellite imagery describes what is happening on the ground. ECO-SCOPE is the translation + evidence layer between the two.

What makes the upgraded version stronger

- It does not assume every EC contains clean coordinates.
- It does not treat OCR as perfect.
- It does not compare arbitrary satellite dates and call every vegetation change a violation.
- It does not pretend 10 m imagery can answer every boundary question.
- It exposes confidence and uncertainty instead of hiding them.
- It distinguishes expected project activity from potentially unexpected activity.
- It keeps a human reviewer in the loop for ambiguous rules and findings.
- It turns one-time analysis into continuous monitoring.
- It creates a feedback loop so verified results can improve the models.

What should be built first

For a hackathon: prove the narrowest possible end-to-end loop. One mining project, a few EC conditions, a resolved boundary, one or two satellite comparisons, one evidence-backed finding, and a reviewer confirmation screen. That is enough to demonstrate the central product insight without pretending the entire 24-month platform already exists.

Positioning

ECO-SCOPE is an automated environmental compliance **screening** and monitoring platform — not a legal proof engine. The source specification explicitly builds this limitation into the product design.

Source note

This redesigned guide is based on the uploaded “ECO-SCOPE — Environmental Compliance Intelligence Platform: Full Problem Analysis, Loophole Resolution, Upgrade Plan & 24-Month Build Roadmap” (22 pages). It reorganizes and simplifies the source for easier reading while retaining its central architecture, proposed upgrades, roadmap, risks and KPI targets.